

THE HUMAN FRONTAL LOBES **THIRD EDITION**

FUNCTIONS AND DISORDERS

**EDITED BY BRUCE L. MILLER
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CHAPTER 1

The Human Frontal Lobes

An Introduction

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Recent research into the functions of the frontal lobes is transforming our understanding of this important brain region. This chapter offers an overview of the frontal lobes and introduces the topics discussed in greater detail throughout this book. A historical review emphasizes the key developments in frontal lobe research generated over the past two centuries. Also, it describes new imaging techniques that play an increasingly central role in the study of the frontal lobes; recent cognitive, social, and pharmacological approaches to frontal lobe functions; and the human diseases in which frontal lobes or frontal-subcortical connections to the frontal lobes are particularly vulnerable.

Historical Overview of Clinical Advances

During the 19th century, John Harlow's description of the social changes in Phineas Gage following traumatic injury to the orbitofrontal cortex (Macmillan, 2001) and Broca's delineation of the language functions of dominant frontal cortex (Berker, Berker, & Smith, 1986) forever changed the way that brain-behavior relationships would be perceived. With the emergence of new histological and anatomical techniques by Cajal, von Economo, and Brodmann, and the growth of

experimental psychology, Tilney suggested in 1928 that the 20th century would become "the century of the frontal lobes." Yet for most of the 20th century, biologically oriented studies of frontal cortex were exceedingly uncommon, and the sequelae of frontal lesions associated with stroke, tumor, trauma, seizures, infections, and neurodegenerative or psychiatric disorders were routinely neglected. Even when procedures such as frontal leukotomy emerged as a routine practice for patients with disabling psychiatric conditions, there was never a systematic attempt to understand the effects of this procedure on cognition or behavior.

In retrospect, this lack of scientific progress had many causes. Undoubtedly, the formal separation of the fields of psychiatry and neurology at the end of the 19th century had a negative influence, leaving higher cortical functions neglected by both fields. For nearly a century, the dominant focus of psychiatry was no longer brain-behavior relationships, whereas neurology became the study of motor and sensory, but not higher cortical, functions. Also, the frontal lobes did not prove easily tractable to clinicians; the cognitive and behavioral deficits associated with frontal injury were not readily apparent or easily characterized. The concepts of executive control and behavioral disorders due to frontal lobe pathology were foreign to the field. Similarly, imaging and

other technologies needed to measure frontal lobe function or structures were lacking. By midcentury, Tilney's predictions appeared premature.

In the 1950s and 1960s, a few investigators began seriously to tackle questions regarding the function of the frontal lobes. In the area of motor systems, Penfield's intraoperative stimulation of patients with neurological disease facilitated an understanding of the functional organization of motor and premotor cortex. Penfield (1954) conceptualized the frontal lobes as having three major anatomical divisions: a motor strip involved with fine coordination of movement; a premotor area devoted to the overall organization of movement; and the mysterious prefrontal region. Stimulation of the motor strip resulted in simple movements of the affected muscles and muscle groups, whereas complex organized movements were seen with premotor stimulation. Yet Penfield seemed to run into a brick wall when he electrically stimulated prefrontal cortex. These prefrontal areas were "silent," and Penfield was unable to explain the functions of these regions on the basis of stimulation studies.

Paul Yakovlev's anatomical studies of the frontal lobes and their connections strongly influenced neurologists and pathologists, including Norman Geschwind, D. Frank Benson, and Arne Brun, who went on to do important research into frontal lobe functions. Yakovlev (1968) focused these emerging investigators on the phylogenetic and developmental origins of frontal cortex and the subcortical-limbic connections to the frontal regions. His work encouraged important new studies that explored the clinical syndromes associated with frontal injury from stroke, trauma, tumor, and neurodegeneration. In addition, Alexander Luria and D. Frank Benson began to focus their research on the vast space of the frontal lobes. Luria (1970), the innovative Russian psychologist, developed novel (and, even by today's standards, remarkably modern) theories about frontal lobe functions. His theories about frontal lobe organization were highly theoretical, but many components of Luria's ideas were shaped and sharpened by his clinical experiences with patients who had focal lesions.

The advent of behavioral neurology under the leadership of Norman Geschwind and D. Frank Benson stimulated a generation of

neurologists and psychologists to explore the anatomical basis for cognition and behavior. Benson, a student and colleague of Geschwind, was particularly intrigued by frontal lobe function and even traveled to Russia to observe Luria's patient-based approach (Benson, 1996). Benson liked to tell how Luria's patients and their families approached him with profound formality, respecting his place in Russia's rigid hierarchy. Patients with frontal lobe injury were different, often greeting Luria like an old friend, entering inappropriately into his personal space.

Benson realized that the frontal lobes had a strong connection to social cognition, and he was one of the first investigators to include the study of behavior in the evaluation of frontal lobe function. The importance of his studies and their influence on his students cannot be underemphasized. Benson stressed the importance of bedside observation, believing that capturing phenomenology was the first step in understanding frontal lobe function, and that theories were greatly bolstered if they followed and were based upon careful clinical observations. Among the behavioral neurologists of his generation, Benson was most willing to explore the interface between neurology and psychiatry, and he realized that the frontal lobes play an important role in the genesis of behaviors that were then considered psychiatric in origin.

Some of Benson's most original work came from his study of the role of the frontal cortex in disorders that were considered psychiatric: syndromes such as reduplicative paramnesia and confabulation associated with amnesia. He hypothesized that a possible mechanism for delusions might be the patient's altered monitoring of distorted or incomplete information (Mercer, Wapner, Gardner, & Benson, 1977). In reduplicative paramnesia, the combination of distorted visual information from a posterior nondominant temporoparietal lesion and abnormal monitoring due to a frontal lesion might lead to delusions about the identity of a place (Benson, Gardner, & Meadows, 1976). In confabulation associated with amnesia, frontal injury might lead a patient to fill in the amnesia with false information, whereas without the frontal injury, the confabulation would not occur (Benson et al., 1996; Stuss & Benson, 1986).

Benson and his colleague and friend Donald Stuss evaluated the effects of leukotomy on

frontal lobe function (Stuss et al., 1981). Most patients with leukotomy were never studied, and Stuss and Benson found patients in whom the procedure had been performed years or even decades earlier. This search represented the beginning of a new era in the study of frontal lobe functions with modern cognitive approaches, sometimes supplemented by neuroimaging. Stuss and Benson eventually modified Luria's classification by suggesting that the prefrontal cortex has three major anatomical divisions: orbitofrontal, cingulate, and dorsolateral. They further proposed that whereas the orbitofrontal cortex modulates social control, and the cingulate cortex is responsible for the generation of goal-directed behavior, many of the neuropsychological functions of the frontal cortex are localized within dorsolateral prefrontal regions (Stuss & Benson, 1986). These concepts have proven remarkably durable across many approaches, whether lesion-based or studied in healthy controls with functional magnetic resonance imaging (fMRI).

Others probed the frontal lobes from novel perspectives. The concept of working memory pioneered by Alan Baddeley (2003) began the parcellation of specific frontal lobe neuropsychological functions to different regions in the frontal cortex. Patricia Goldman-Rakic and colleagues (e.g., Chafee & Goldman-Rakic, 1998) emphasized the role of a frontoparietal system in visual attention/working memory tasks, and showed the role and anatomy of the dopaminergic components of this circuit. Mark D'Esposito and colleagues (Ranganath & D'Esposito, 2005; D'Esposito & Postle, 2015), and others, have used functional imaging techniques such as fMRI to evaluate this working memory system *in vivo*. Marsel Mesulam's (1998) histological and phylogenetic approaches to cortical organization have emphasized the functional organization of cortex based on connections to primary sensory or motor areas. Edmund Rolls's (2004) pioneering research with single-cell recording in rodents demonstrated the importance of orbitofrontal cortex in reward and eating-related behaviors. Finally, Antonio Damasio's (2003) "somatic marker" theory suggested that the orbitofrontal cortex is involved with the interpretation of autonomic information in decision making. More recent work from Levenson, Sturm, and Haase (2014) has delineated an important role for the anterior cingulate

cortex as a hub for the interpretation of social data that is also linked to autonomic output.

Imaging

Neuroimaging research continues as the pre-eminent technique for evaluation of the frontal lobes. Computed tomography (CT) became widely available in the late 1970s; before that, clinicians had few ways to visualize the structure or function of frontal cortex. Older tools like pneumoencephalography and angiography were highly invasive, painful, and sometimes dangerous, and offered only crude outlines of frontal anatomy. With CT, suddenly clinical-anatomical correlations could be made *in vivo*, and the presence and extent of structural lesions associated with stroke, demyelination, tumor, or trauma became visible before death. CT also allowed researchers to explore the relationship between volume of frontal injury and clinical status with trauma, tumor, stroke, and neurodegeneration. Measurement of atrophy became a way to differentiate normal aging from dementia and to separate the distinctive dementia syndromes from each other. Yet CT had constraints related to the frontal lobes. In particular, artifacts generated from adjacent bone diminished visualization of orbitofrontal and anterior frontal structures. Also, resolution between gray and white matter was limited.

Development of MRI in the early 1980s circumvented many of the problems associated with CT. Better resolution of gray and white matter, and disappearance of bone-related artifacts, made MRI the technique of choice for the evaluation of the frontal lobes. MRI allowed quantitative measurement of frontal volumes, facilitating studies on the role of frontal cortex or subfrontal white matter lesions in the pathogenesis of a wide variety of cognitive and behavioral syndromes.

Volumetric region-of-interest (ROI) measures of normal or pathological tissue, correlated with specific cognitive or behavioral parameters, remain a widely used approach to the study of sequelae from frontal injury. Hand-drawn ROIs in different frontal regions have facilitated the study of neurological and psychiatric disorders. Because this approach is slow, and interrater reliability is a methodological concern, automatic ROI methods are being developed for measuring the frontal

lobes in their entirety and for selective regional analyses.

Other MRI-based techniques have facilitated study of brain–behavior relationships related to the frontal lobes, including voxel-based morphometry (VBM) and deformation–tensor morphometry (DTM). With these techniques, the overall patterns of brain atrophy can be compared between two groups. Often a disease cohort is compared to age-matched controls, and VBM and DTM allow comparisons between an individual patient and a control group (Ashburner & Friston, 2001; Studholme et al., 2004). Or a patient's results can be compared with use of images obtained at different times. Unlike ROI approaches, VBM and DTM require no a priori hypotheses and allow comparisons of the whole brain (or specific frontal regions) between patient cohorts and controls. These techniques have proven particularly powerful for delineating the most vulnerable brain regions in different conditions that cause dementia. One exciting development has been the capability of performing multimodal imaging analyses of individual patients (Zhang et al., 2011).

Structural MRI still holds great promise for facilitating a better understanding of the frontal lobes. VBM and ROI approaches are still relatively new, and studies combining these techniques with emerging cognitive approaches should offer many new insights into the structural and functional relationships in the frontal lobes. Movement from 1.5- to 3.0-tesla (T) magnets has improved resolution of frontal regions and has facilitated better separation of gray matter from white matter. Furthermore, more powerful magnets (7-T magnets) will offer even greater advantages for structural resolution of the frontal lobes in the coming decade.

Functional methods using radionuclides such as single-photon emission computed tomography (SPECT) and positron emission tomography (PET) have contributed to a better understanding of the frontal lobes. SPECT and PET have helped to show the frontal component of a wide variety of disease states, including depression (Mayberg, 2002; Holtzheimer & Mayberg, 2011), obsessive–compulsive disorder (Saxena et al., 2004), schizophrenia (Choudhary et al., 2015), attention-deficit/hyperactivity disorder (ADHD) (Schweitzer et al., 2003), Alzheimer's disease (AD) (Craig et al., 1996), and frontotemporal dementia

(FTD) (Miller et al., 1991). Similarly, frontal contributions to working memory, executive function, apathy, and disinhibition continue to be explored with these functional techniques.

Increasingly, MRI is being explored as a way to supplement, or even replace, SPECT and PET techniques for measuring brain function. Perfusion MRI (Callen, Black, Caldwell, & Grady, 2004) offers better resolution and is less invasive than either SPECT or PET, but systematic comparisons among perfusion MRI, SPECT, and PET are still lacking. fMRI is a new and powerful tool that has dramatically changed research into the frontal lobes by allowing noninvasive evaluation *in vivo*. Relying upon changes in the MRI signal that occur when the brain undergoes metabolic activation (the BOLD signal), fMRI allows study of brain activity associated with specific cognitive tasks (Frackowiak, 2000–2001). Frontal brain regions that are active during working memory, self-reflection, word generation, temporal sequencing, set shifting, and many other cognitive paradigms are being evaluated by psychologists, psychiatrists, and neurologists with this technique (Baron-Cohen, 2004). Another exciting development with fMRI has been the mapping of networks activated with cognitive tasks or during rest (Greicius, Krasnow, Reiss, & Menon, 2003). This elucidates an understanding of which brain systems work together during specific activities. No technique has ever generated quite so many new data regarding the functional organization of the frontal lobes, and it will be many years before these data can be fully understood.

Magnetoencephalography, transcranial magnetic stimulation, and nuclear magnetic resonance spectroscopy represent other imaging approaches that have been applied to the study of frontal lobe functions in healthy or disease states. Whatever the technique, it is clear that neuroimaging will continue to influence our understanding of the frontal lobes.

Neuroanatomical/Functional Organization

The frontal lobes are no longer considered a single functional entity. Rather, there are now various ways of anatomically subdividing this brain region, all based upon distinctive constructs. Most researchers accept that the frontal lobes have three major divisions: motor,

premotor, and prefrontal regions. Motor and premotor areas are considered distinctive functional units, whereas prefrontal cortex is more complex, requiring further subdivision. One system to subdivide the frontal cortex relies on the distinctive functions of different prefrontal regions. Another approach considers regional connections to and from specific subcortical regions. In addition, the left and right frontal lobes are increasingly differentiated: The left frontal lobe is more specialized for language-related functions, and the right frontal region is dominant in social cognition and emotion. Analyzing regional histology represents yet another way to subdivide the frontal lobes.

A functional approach suggested by Cummings (1985) and Stuss and Benson (1986) divides prefrontal cortex into orbital, dorsolateral, and cingulate regions. Increasingly, these gross divisions are being further parcellated into smaller functional units. Baddeley (2003) and others have shown a dorsal prefrontal (Brodmann's area [BA] 46) parietal system involved with working memory (left, verbal working memory; right, visual working memory). A more ventral frontoparietal system is involved with mirror movements (Rizzolatti, Fogassi, & Gallese, 2002), speech initiation (Dronkers, 1996), and affect matching (Rosen et al., 2004). Orbitofrontal cortex has important medial-lateral and right-left divisions. Medial orbitofrontal cortex is strongly connected with hypothalamic nuclei, whereas lateral orbital cortex is more strongly connected with anterior temporal and insular regions. In certain behavioral paradigms, orbitofrontal regions show antagonistic functions. For example, the medial area is activated when an individual is hungry, whereas the lateral region is activated when the same individual is sated (Small, Zatorre, Dagher, Evans, & Jones-Gotman, 2001).

"Executive control," a broad term used to describe the neuropsychological functions of the frontal lobes, incorporates many distinctive cognitive processes. Neuropsychologists are now attempting to subdivide neuropsychological functions into their anatomically driven subcomponents. Working memory is a core constituent of executive functioning, and when there are deficits in working memory, nearly all other tests of executive control are vulnerable. Other components of executive control include generation, inhibition, set shifting, concept formation, temporal

sequencing, insight, interpersonal perspective taking (theory of mind), and social and real-world executive performance. Distinctive tasks that capture these different aspects of executive control are in development. Similarly, the anatomical components of these tasks are under investigation. This topic is explored by Staffaroni, Stephens, and Kramer (Chapter 8, this volume). Left versus right frontal lobe functions are now under careful scrutiny as well. Linguistic functions of Broca's area in prefrontal cortex have been understood for nearly 150 years, but only fairly recently have investigators begun formally addressing the preeminence of the right prefrontal cortex for social and emotional behavior. For example, in an imaging study of a large cohort of patients with dementia, Rosen and colleagues (2004) found that repetitive compulsive behaviors were strongly associated with right supplementary motor area atrophy, disinhibition with atrophy in the right orbitofrontal cortex, and apathy with atrophy in the right cingulate cortex. Right insular atrophy was correlated with abnormalities in eating. fMRI approaches to behavior suggest that self-reflection, reading the emotional expressions of others, and many other aspects of social cognition activate more right than left frontal regions. The specifics of behavioral specialization of right frontal cortex remain poorly understood, but the combination of fMRI and lesion studies should elucidate many of these details in the coming decades.

Yakovlev (1968), and more recently Cummings (1993), Tekin and Cummings (2002), Flaherty and Graybiel (1995), and Lee et al. (2014), have emphasized the intimate connections between subcortical and frontal structures, and this connectivity has helped to organize the frontal lobes into distinctive units. The presence of these anatomical circuits suggests one way to classify the organization of the frontal lobes. Cummings (1993) suggests that there are five distinctive frontal-subcortical systems: (1) supplementary motor area, (2) frontal eye fields, (3) dorsolateral prefrontal cortex, (4) orbitofrontal cortex, and (5) anterior cingulate cortex. Within each circuit, specific neurochemical systems transmit their functional activity. Lesions in these different frontal or subcortical circuits lead to distinctive clinical syndromes. With supplementary motor dysfunction, deficits in controlled movement and repetitive motor behaviors

emerge (Gorno-Tempini, Murray, Rankin, Weiner, & Miller, 2004); with eye movement injury, control of gaze is diminished (Chou & Lisberger, 2004); with dorsolateral dysfunction, there is loss of executive control and neuropsychological deficits in the area of working memory, and alternation and planning deficits emerge (Boone et al., 1999); orbitofrontal dysfunction causes social deficits, including disinhibition with sparing of cognition (Damasio, 2003); and cingulate lesions lead to “amotivational” states (Tekin & Cummings, 2002). Psychiatric syndromes are strongly linked to these frontal–subcortical circuits, particularly depression, mania, and obsessive–compulsive disorder. Recent work by Lee and colleagues (2014) suggests that thalamic dysfunction leads to frontal behavioral disorders in a subtype of FTD associated with mutations in the *C9ORF* gene. The relationship of these psychiatric syndromes to frontal lobe dysfunction is discussed extensively in this book. Finally, movement disorders are particularly prominent when the basal ganglia component of frontal–subcortical circuits is affected.

Mesulam (1998) considers anatomical connectivity and histology in his approach to cortical organization of the frontal regions. His system emphasizes limbic, paralimbic, and cortical divisions. Cortical regions are divided into four subtypes—primary, unimodal, heteromodal, and supramodal cortex—based on the nature of these regions’ other cortical connections. With this system, the first-order regions include primary motor (BA 4), sensory (BA 3), visual (BA 17), auditory (BA 42), and gustatory (BA 43) cortex. These regions are theoretically bound together by the fact that they are the first cortical regions to connect with subcortical areas that translate information from or to the external milieu. Unimodal regions have efferent connections from the primary areas and process only one type of sensory stimuli (visual, auditory, or gustatory). Heteromodal regions are areas that connect with unimodal areas and are involved with multiple types of sensory processing. In the final, supramodal division, many prefrontal regions are included because they are remote from sensory processing and involved with higher-order processing.

Finally, the role of a histologically unique neuron, the spindle cell, is being explored in relation to the distinctive functions of the frontal lobes in humans. Originally described by von Economo, these large, spindle-shaped

cells found in layer 5b of the ventral frontal anterior cingulate and anterior insular regions are more strongly localized to the right frontal area. These cells first appeared in great apes, but are abundant only in humans. They first appeared in orangutans, but are far more numerous in gorillas. Spindle cells show still greater abundance in bonobo chimpanzees and have 1,000-fold greater concentration in humans than in chimps. Spindle cells are absent in gibbons, New World monkeys, Old World monkeys, and all other mammals studied (Allman, Hakeem, & Watson, 2002; Nimchinsky et al., 1999). The role of these cells in social-cognitive paradigms will be extensively studied in the coming decade.

Neurological Disorders of the Frontal Lobes

One important approach to understanding frontal function is to examine neurological disorders that selectively attack the frontal regions. Neuroimaging has facilitated this approach by allowing investigators to outline brain areas affected by distinctive pathological processes. Many neurological disorders involve the frontal lobes either directly or through frontal–subcortical connections. AD is an example of a disease in which frontal cortex is not the primary site of injury, but is usually affected at some time in the course of the illness (Ossenkoppele et al., 2015). With AD, the combination of imaging, behavioral, and cognitive approaches has helped to clarify the role of the cingulate cortex in a variety of psychological processes, including apathy and compulsions. There are also a few conditions in which frontal cortex is the primary site of disease. Frontotemporal lobar degeneration (FTLD) is one such example, and this disorder has become a powerful model for the investigation of frontal function. Three subtypes of FTLD exist, all defined by distinctive clinical syndromes (Rosen et al., 2002). The core features of the major subtype, FTD, include loss of social and personal conduct, loss of insight, and emotional blunting. The most profound atrophy with FTD is in the anterior cingulate, insular, and orbitofrontal cortices. The right hemisphere is more severely affected than the left. In the second subtype, progressive nonfluent aphasia, the atrophy involves these same frontal regions, but is more severe on the left than on the right side. With semantic dementia,

the third FTLT subtype, the insula and orbitofrontal cortex, amygdala, and anterior temporal lobes are bilaterally atrophic. These distinctive anatomies facilitate exploration of the functions of the right versus left frontal cortex and different frontal regions. Each patient with FTLT shows slightly distinctive involvement of dorsolateral, orbital, cingulate, or insular regions. This allows comparison of individuals, or groups of patients, with unique anatomical patterns, with the goal of parceling out the specific functions of these frontal regions.

There are other neurological disease models for the evaluation of frontal lobe function. Beginning with Harlow's descriptions of Phineas Gage, trauma has helped to delineate the functional anatomical divisions in the frontal lobes. Often trauma selectively injures orbitofrontal cortex while sparing dorsolateral regions (Alexander & Stuss, 2000). In patients with these selective injuries, profound deficits in social behavior and intelligence, with sparing of the cognitive components of frontal lobe function, led to the recognition that orbitofrontal and dorsolateral components of the frontal lobes are distinctive. Similarly, from the clinical study of patients with selective injury to the cingulate cortex associated with anterior cerebral artery stroke or psychosurgery (Brower & Price, 2001), researchers now suspect that this area is particularly important for initiation of behavior.

Psychiatric Disorders and the Frontal Lobe

An important paradigm shift in psychiatry has been the recognition of dysfunction in the prefrontal cortex or within frontal-subcortical circuitry in many psychiatric disturbances. This conceptual shift has been driven by converging evidence generated through a wide variety of approaches: cognitive, behavioral, neuroimaging, and neuropathological. Schizophrenia represents one such disorder in which all of these distinctive parameters suggest frontal dysfunction.

Neuropsychological testing of many patients with schizophrenia shows deficits in frontal/executive skills compared to those of controls (Kremen, Seidman, Faraone, Toomey, & Tsuang, 2004). Similarly, the behavioral features of patients with schizophrenia have many parallels to those of patients with

frontal injury. The negative symptom complex found in many patients with schizophrenia is similar to the apathetic syndromes caused by frontal injury (Boone, Miller, Swartz, Lu, & Lee, 2003). Furthermore, patients with schizophrenia who have the most severe negative symptoms, such as apathy, tend to have the smallest frontal brain volumes, further linking specific symptoms in schizophrenia to the frontal lobes (Roth, Flashman, Saykin, McAllister, & Vidaver, 2004). Also, a wide variety of PET and fMRI studies show metabolic abnormalities at rest and during activation in prefrontal cortex (Reckless, Andreassen, Server, Østefjells, & Jensen, 2015). Finally, autopsies of many patients with schizophrenia reveal neuropathological abnormalities. Most of these changes are found in the hippocampus, but recent work suggests abnormalities in the frontal regions as well (Webster, O'Grady, Kleinman, & Weickert, 2005; Balla & Frescka, 2011).

Schizophrenia represents just one of the many psychiatric illnesses in which behavioral disturbance is linked to the frontal lobes. Major depression is associated with hypometabolism in the dorsal and orbitofrontal cortex (Mayberg, 2002). Metabolic changes have also been noted in the anterior cingulate and insular regions. Similarly, disproportionate atrophy in the orbitofrontal cortex is seen in elderly patients with depression or bipolar disorder. Also, injury to frontal cortex is often followed by depressive syndromes, and right orbitofrontal injury can trigger secondary manic syndromes. Frontal-subcortical circuits have also been hypothesized to participate in obsessive-compulsive disorder, and this hypothesis is supported by both structural and functional imaging. Finally, ADHD is clearly associated with dysfunction in the prefrontal cortical, attentional, and executive systems.

Summary

The frontal lobes constitute a large brain region representing 30% of the cortical surface. The coming decades will see more definitive explorations of the structural, functional, neurochemical, and histological underpinnings of the frontal lobes. Understanding the frontal lobes will lead to advances against a wide variety of neurological and psychiatric conditions. These new developments are outlined throughout this book.

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